

SOLOMON BAYE

INFORMATION SECURITY ARCHITECT

Enterprise Security Architecture | IAM & Zero Trust | Application & API Security | Cloud | Telecom & FinTech
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PROFESSIONAL PROFILE

Information Security Architect with 10 years in telecommunications, including 6 years in information security architecture at Ethio Telecom, Ethiopia's national telecom operator with 100 million customers. Designs and assures secure architectures for digital financial services, identity and access management, Zero Trust, application and API integration, and multi-cloud environments. Translates business strategy and risk into security services, control objectives, governance models and roadmaps using SABSA, ArchiMate and TOGAF-aligned practice. CISA certified, IEEE-published researcher and conference speaker.

10 yrs telecom | 6 yrs security architecture | 50,000+ identities protected | 15+ platforms assured | 40% assessed risk reduction

**Contributed to People Process Technology initiatives associated with a 40% reduction in assessed enterprise risk.*

PROFESSIONAL EXPERIENCE

Information Security Architect

2020 – Present

Ethio Telecom | Addis Ababa, Ethiopia

- **telebirr Enterprise Security Architecture:** Led security architecture for a digital financial services platform serving millions of customers, using SABSA and ArchiMate to trace business drivers, trust relationships, registration flows and security services through to layered controls.
- **Enterprise IAM:** Architected IAM capabilities protecting 50,000+ identities, including an IBM IAM deployment for 30,000+, covering authentication, authorization, privileged access (CyberArk PAM) and enterprise application integration.
- **Zero Trust:** Designed identity-centric access and Zero Trust controls that extend consistent authentication and authorization across applications, privileged accounts and integrated systems.
- **Cloud Security:** Governed security architecture assurance across AWS, Microsoft Azure and Huawei Cloud, assessing workloads, identity controls, network architecture, security posture and technology risk.
- **Architecture Assurance:** Reviewed and validated third-party designs across 15+ major platforms (IBM IAM, QRadar SIEM/SOAR, CyberArk, Radware WAF/Anti-DDoS, Huawei AAA/VDI/Cloud, WSO2, Palo Alto, Cisco, Allot DPI), authoring HLD, LLD, ICD and Production Acceptance Test documentation.
- **API & Integration Security:** Reviewed API and legacy-integration architectures, identifying application, data and network risks and delivering redesign recommendations and compensating controls.
- **Risk & Governance:** Ran gap assessments and control reviews aligned to NIST CSF, ISO 27001/27005, ISO 31000 and COBIT, contributing to initiatives associated with a 40% reduction in assessed enterprise risk.
- **Advisory & Leadership:** Serve as architecture assurance and audit focal point, informing technology and security investment decisions; author security policies, architecture standards and executive presentations; brief executives, vendors and business teams; deliver security training.

External Security Architecture Consultant

Independent engagement

- Delivered an end-to-end Enterprise Architecture engagement covering business and risk analysis, architecture assessment, solution recommendation and implementation sign-off.

Network Engineer

2018 – 2020

Ethio Telecom | Addis Ababa, Ethiopia

- Engineered and hardened routing, firewall and Wi-Fi access controls supporting 99.9% availability across critical network services; deployed and managed WAF and NGFW platforms.
- Managed MSAG-to-MSAN migrations and FTTH/copper rollouts for 3,000+ customers, coordinating security planning and acceptance with Huawei vendor teams.
- Provided security design input for early enterprise cloud and hybrid network integrations, laying the foundation for the move into security architecture.

Customer Service Consultant

2016 – 2018

Ethio Telecom | Addis Ababa, Ethiopia

- First-line escalation points for complex technical incidents: root-cause analysis and coordinated resolution with engineering teams while maintaining 95%+ customer satisfaction.

SELECTED PROJECT

Security Blueprint

Independent project

Self-developed SABSA enterprise architecture and risk modeling platform

- Designed and built a platform spanning asset discovery, gap assessment, target architecture, architecture assurance and executive reporting, structured on SABSA Layers 1–5 (Contextual to Component).
- Built an assessment matrix linking business drivers, business attributes, security requirements, control objectives, implementation status and control-effectiveness scoring, with ISO 27001 Annex A traceability.
- Added automated recommendations, configurable scopes (Mobile Money, Telecom, Cloud Native, OT/ICS), Word/Excel reporting, maturity scoring, roadmaps and executive dashboards.

RECOGNITION & RESEARCH

- **Speaker, African Enterprise Security Architecture Conference (AEAC 2026):** presented practical approaches to enterprise security architecture, governance and security transformation.
- **IEEE Researcher, SEGE 2026 (Canada):** authored peer-reviewed research on IT/OT cybersecurity convergence, presented at the IEEE International Conference on Smart Energy Systems Engineering. Control Systems MSc background bridges enterprise security with critical infrastructure.
- **IBM IAM Vendor Training, Shanghai, China:** onsite vendor-led IBM IAM training and architecture-focused technical workshop.

TECHNOLOGY & CAPABILITIES

Enterprise Architecture: SABSA, ArchiMate, TOGAF-aligned architecture, target and reference architecture, security services, control objectives, roadmaps

Identity & Zero Trust: IBM IAM, IAM/IGA, CyberArk PAM, Huawei AAA, identity federation, privileged access, Zero Trust

Application & Integration: WSO2 API Management, API security, legacy integration, solution architecture, HLD/LLD/ICD

Cloud: AWS, Microsoft Azure, Huawei Cloud, hybrid cloud, security posture assurance

Cyber Defense: QRadar SIEM/SOAR, Radware WAF/Anti-DDoS, Palo Alto, Cisco, Allot DPI, NGFW, threat modeling

Governance & Risk: NIST CSF, ISO 27001/27005, ISO 31000, COBIT, gap assessment, third-party review, vendor governance

EDUCATION & CREDENTIALS

CISA – Certified Information Systems Auditor

ISACA

MSc, Electrical & Computer Engineering (Control Systems)

Addis Ababa University | 2016 – 2019

BSc, Electrical & Computer Engineering

Addis Ababa University | 2010 – 2015 | Top 5 of class